

COURSE CODE/TITLE:	CPE 212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	23/07/2026	DATE SUBMITTED:	23/07/2026
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ACTIVITY NO. 3.1	3.1 Hands-on Activity - Install SSH server on CentOS or RHEL 8
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1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:

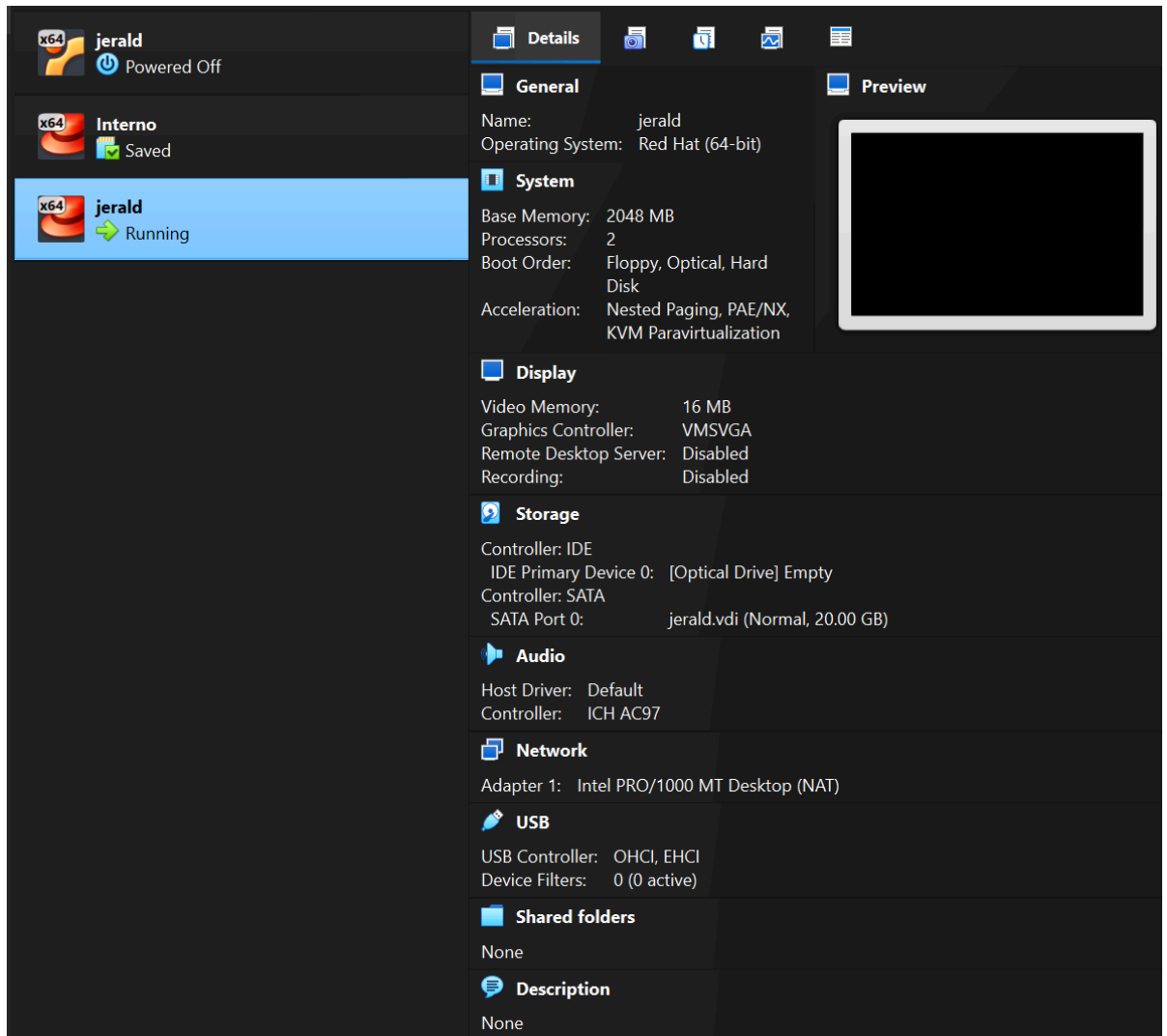
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/

 CentOS-Stream-9-latest-x86_64-dvd1	23/07/2026 11:36 am	Disc Image File	15,202,688 KB
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2. Create a VM machine with 2 Gb RAM and 20 Gb HD.

3. Install the downloaded image.

4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

```
$ dnf install openssh-server
```

```
[root@localhost ~]# dnf install openssh-server
CentOS Stream 9 - BaseOS                1.7 MB/s | 9.0 MB      00:05
CentOS Stream 9 - AppStream              1.7 MB/s | 28 MB       00:16
CentOS Stream 9 - Extras packages        18 kB/s | 21 kB        00:01
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Start the sshd daemon and set to start after reboot:

```
$ systemctl start sshd
```

```
$ systemctl enable sshd
```

```
[root@localhost ~]# systemctl start sshd
[root@localhost ~]# systemctl enable sshd
```

3. Confirm that the sshd daemon is up and running:

\$ systemctl status sshd

```
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
   Active: active (running) since Thu 2026-07-23 22:54:33 PST; 10min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 966 (sshd)
      Tasks: 1 (limit: 10500)
     Memory: 1.7M (peak: 1.9M)
        CPU: 32ms
    CGroup: /system.slice/sshd.service
            └─966 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 22:54:33 localhost.localdomain systemd[1]: Starting OpenSSH server daemon>
Jul 23 22:54:33 localhost.localdomain sshd[966]: Server listening on 0.0.0.0 po>
Jul 23 22:54:33 localhost.localdomain sshd[966]: Server listening on :: port 22.
Jul 23 22:54:33 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

4. Open the SSH port 22 to allow incoming traffic:

\$ firewall-cmd --zone=public --permanent --add-service=ssh

\$ firewall-cmd --reload

```
[root@localhost ~]# firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@localhost ~]# firewall-cmd --reload
success
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

\$ systemctl reload sshd

```
[root@localhost ~]# nano /etc/ssh/sshd_config
[root@localhost ~]# systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.

```
[root@localhost ~]# ssh -V
OpenSSH_9.9p1, OpenSSL 3.5.7 9 Jun 2026
```

2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
[root@localhost ~]# ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase for "/root/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:lFEXebgTD0xP5SU2InI04UL0uHiJyGKUoWiGxvLRKs root@localhost.localdomain
The key's randomart image is:
+---[RSA 4096]-----+
|  . . . =oB+=+=.o |
| +..+.. . 0 +=+.+ |
| ++o.o   = o  *. . |
| +..o.. + +  o .   |
|   +.o o S      .   |
| E . . .           |
|-----+
+-----[SHA256]-----+
```

3. On CentOS, verify that you have the authorized_keys.

```
[root@localhost ~]# ssh-copy-id jeraldinterno@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
jeraldinterno@192.168.56.101's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'jeraldinterno@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
[root@localhost ~]# ssh jeraldinterno@192.168.56.101
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 22:57:00 2026
```

2. Show evidence that you are connected.

```
[jeraldinterno@localhost ~]$ hostnamectl
  Static hostname: (unset)
  Transient hostname: localhost
    Icon name: computer-vm
      Chassis: vm
    Machine ID: 8401b194b9c84dc894fba2f2c7237ae5
      Boot ID: f91155a5322f499ca7405294a7e76fb7
  Virtualization: oracle
  Operating System: CentOS Stream 9
    CPE OS Name: cpe:/o:centos:centos:9
      Kernel: Linux 5.14.0-725.el9.x86_64
    Architecture: x86-64
  Hardware Vendor: innotek GmbH
    Hardware Model: VirtualBox
  Firmware Version: VirtualBox
```

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- When choosing between Debian and Red Hat, consider whether you need a free, community-driven system focused on long-term stability or an enterprise-grade platform backed by commercial support. You should also look at package management preferences (apt vs. dnf), software release cycles, and whether your goal is running personal projects or gaining skills for enterprise corporate environments.

2. What are the main differences between Debian and Red Hat Linux distributions?

- The main difference is that Debian is a completely free, community-maintained Linux distribution using .deb packages and apt, whereas Red Hat Enterprise Linux (RHEL) is a commercial distribution backed by Red Hat/IBM that uses .rpm packages, dnf/yum, and paid enterprise support. Additionally, Debian prioritizes non-commercial software freedom and community-driven stability, while Red Hat focuses on corporate integration, compliance, and paid service agreements.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code assistance and explanation of concepts

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of



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concepts, etc.

Extent of Use:

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